

Principal Component Analysis (PCA) on Boston Housing Dataset

Set Working Directory

```
setwd("/Users/Ian/Documents/Code_Projects/boston-house-price-modeling/descriptive_analysis")
getwd()
```

```
## [1] "/Users/Ian/Documents/Code_Projects/boston-house-price-modeling/descriptive_analysis"
```

1. Load Libraries

```
library(ggplot2)
library(dplyr)
library(ggrepel)  # for non-overlapping labels
library(patchwork)
```

2. Load Dataset

```
# Read the cleaned dataset (assumes 'B' column removed)
data <- read.csv("../data/boston_cleaned.csv")
str(data)
```

```
## 'data.frame':  506 obs. of  13 variables:
## $ CRIM   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ ZN     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ INDUS  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ CHAS   : int   0 0 0 0 0 0 0 0 0 0 ...
## $ NOX    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
## $ RM     : num  6.58 6.42 7.18 7 7.15 ...
## $ AGE    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ DIS    : num  4.09 4.97 4.97 6.06 6.06 ...
## $ RAD    : int   1 2 2 3 3 3 5 5 5 5 ...
## $ TAX    : int  296 242 242 222 222 222 311 311 311 311 ...
## $ PTRATIO: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ LSTAT  : num  4.98 9.14 4.03 2.94 5.33 ...
## $ MEDV   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

3. Basic Statistics and Correlation

```
# Correlation matrix
cor(data)
```

```
##          CRIM          ZN          INDUS          CHAS          NOX
## CRIM      1.00000000 -0.20046922  0.40658341 -0.055891582  0.42097171
## ZN        -0.20046922  1.00000000 -0.53382819 -0.042696719 -0.51660371
## INDUS     0.40658341 -0.53382819  1.00000000  0.062938027  0.76365145
## CHAS     -0.05589158 -0.04269672  0.06293803  1.000000000  0.09120281
## NOX       0.42097171 -0.51660371  0.76365145  0.091202807  1.00000000
## RM       -0.21924670  0.31199059 -0.39167585  0.091251225 -0.30218819
## AGE       0.35273425 -0.56953734  0.64477851  0.086517774  0.73147010
## DIS      -0.37967009  0.66440822 -0.70802699 -0.099175780 -0.76923011
## RAD       0.62550515 -0.31194783  0.59512927 -0.007368241  0.61144056
## TAX       0.58276431 -0.31456332  0.72076018 -0.035586518  0.66802320
## PTRATIO  0.28994558 -0.39167855  0.38324756 -0.121515174  0.18893268
## LSTAT     0.45562148 -0.41299457  0.60379972 -0.053929298  0.59087892
## MEDV     -0.38830461  0.36044534 -0.48372516  0.175260177 -0.42732077
##          RM          AGE          DIS          RAD          TAX          PTRATIO
## CRIM     -0.21924670  0.35273425 -0.37967009  0.625505145  0.58276431  0.28994558
## ZN       0.31199059 -0.56953734  0.66440822 -0.311947826 -0.31456332 -0.39167855
## INDUS    -0.39167585  0.64477851 -0.70802699  0.595129275  0.72076018  0.38324756
## CHAS     0.09125123  0.08651777 -0.09917578 -0.007368241 -0.03558652 -0.1215152
## NOX     -0.30218819  0.73147010 -0.76923011  0.611440563  0.66802320  0.1889327
## RM       1.00000000 -0.24026493  0.20524621 -0.209846668 -0.29204783 -0.3555015
## AGE     -0.24026493  1.00000000 -0.74788054  0.456022452  0.50645559  0.2615150
## DIS     0.20524621 -0.74788054  1.00000000 -0.494587930 -0.53443158 -0.2324705
## RAD     -0.20984667  0.45602245 -0.49458793  1.000000000  0.91022819  0.4647412
## TAX     -0.29204783  0.50645559 -0.53443158  0.910228189  1.00000000  0.4608530
## PTRATIO -0.35550149  0.26151501 -0.23247054  0.464741179  0.46085304  1.0000000
## LSTAT   -0.61380827  0.60233853 -0.49699583  0.488676335  0.54399341  0.3740443
## MEDV     0.69535995 -0.37695457  0.24992873 -0.381626231 -0.46853593 -0.5077867
##          LSTAT          MEDV
## CRIM      0.4556215 -0.3883046
## ZN        -0.4129946  0.3604453
## INDUS     0.6037997 -0.4837252
## CHAS     -0.0539293  0.1752602
## NOX       0.5908789 -0.4273208
## RM       -0.6138083  0.6953599
## AGE       0.6023385 -0.3769546
## DIS     -0.4969958  0.2499287
## RAD       0.4886763 -0.3816262
## TAX       0.5439934 -0.4685359
## PTRATIO  0.3740443 -0.5077867
## LSTAT     1.0000000 -0.7376627
## MEDV     -0.7376627  1.0000000
```

4. PCA Analysis

```
# Run PCA on predictors only (exclude MEDV)
pr.out <- prcomp(data %>% select(-MEDV), scale. = TRUE)
```

4.1 PCA Object Details

```
# Structure of the PCA object  
str(pr.out)
```

```
## List of 5  
## $ sdev      : num [1:12] 2.43 1.183 1.087 0.924 0.896 ...  
## $ rotation: num [1:12, 1:12] 0.2513 -0.26638 0.35487 0.00759 0.34995 ...  
## ..- attr(*, "dimnames")=List of 2  
## .. ..$ : chr [1:12] "CRIM" "ZN" "INDUS" "CHAS" ...  
## .. ..$ : chr [1:12] "PC1" "PC2" "PC3" "PC4" ...  
## $ center   : Named num [1:12] 3.6135 11.3636 11.1368 0.0692 0.5547 ...  
## ..- attr(*, "names")= chr [1:12] "CRIM" "ZN" "INDUS" "CHAS" ...  
## $ scale    : Named num [1:12] 8.602 23.322 6.86 0.254 0.116 ...  
## ..- attr(*, "names")= chr [1:12] "CRIM" "ZN" "INDUS" "CHAS" ...  
## $ x        : num [1:506, 1:12] -2.05 -1.39 -2.03 -2.58 -2.41 ...  
## ..- attr(*, "dimnames")=List of 2  
## .. ..$ : NULL  
## .. ..$ : chr [1:12] "PC1" "PC2" "PC3" "PC4" ...  
## - attr(*, "class")= chr "prcomp"
```

```
# Means and standard deviations  
pr.out$center
```

	CRIM	ZN	INDUS	CHAS	NOX	RM
##	3.61352356	11.36363636	11.13677866	0.06916996	0.55469506	6.28463439
##	AGE	DIS	RAD	TAX	PTRATIO	LSTAT
##	68.57490119	3.79504269	9.54940711	408.23715415	18.45553360	12.65306324

```
pr.out$scale
```

	CRIM	ZN	INDUS	CHAS	NOX	RM
##	8.6015451	23.3224530	6.8603529	0.2539940	0.1158777	0.7026171
##	AGE	DIS	RAD	TAX	PTRATIO	LSTAT
##	28.1488614	2.1057101	8.7072594	168.5371161	2.1649455	7.1410615

4.2 Loadings (Eigenvectors)

```
# Loadings of original variables on PCs  
pr.out$rotation
```

	PC1	PC2	PC3	PC4	PC5
## CRIM	0.251296229	-0.27355692	0.35136928	-0.05982119	0.19174360
## ZN	-0.266381728	-0.25012639	0.35927456	-0.19418290	0.40189844
## INDUS	0.354873085	0.09314825	-0.01487534	-0.01066292	0.01165547
## CHAS	0.007588523	0.50285462	0.20141592	-0.81123189	-0.19730909
## NOX	0.349948498	0.23250734	0.07843338	0.08986757	0.21548522
## RM	-0.196439268	0.27304276	0.56144366	0.40156629	-0.28511864
## AGE	0.323177006	0.29297560	-0.07545447	0.16295868	0.07629772

```
## DIS      -0.330882488 -0.34262077  0.01326386 -0.23446865  0.02963260
## RAD       0.322249962 -0.23092958  0.40774884 -0.04568292 -0.11883781
## TAX       0.342255662 -0.21304270  0.33125764 -0.03986097 -0.03749351
## PTRATIO  0.210942091 -0.39255695 -0.18370909 -0.10866039 -0.70201318
## LSTAT     0.315339517 -0.12810818 -0.26399228 -0.18481855  0.34622055
##          PC6          PC7          PC8          PC9          PC10
## CRIM     -0.76048422  0.15496235 -0.27154624  0.027896076 -0.10919148
## ZN        0.29478619 -0.40141425 -0.37418340  0.259110165  0.26781095
## INDUS     0.34531635  0.17423644 -0.63342361 -0.373501844 -0.31344343
## CHAS     -0.08317501 -0.02508995  0.01730740 -0.011827903  0.01462322
## NOX       0.20787477  0.08040938  0.01401728  0.203177440  0.13622402
## RM       -0.06855657 -0.32731773 -0.02827600 -0.440521470  0.04552539
## AGE      -0.10894852 -0.59993427  0.04471045  0.391517675 -0.45565391
## DIS       0.09997861 -0.12237288  0.16231933 -0.166048521 -0.69009919
## RAD       0.14902205  0.08077154  0.46235895  0.056560825  0.03032156
## TAX       0.32930065  0.08344957  0.16789962 -0.002099586 -0.11450010
## PTRATIO -0.01277945 -0.31806844 -0.25503744  0.125667602  0.18553131
## LSTAT    -0.08953306 -0.42510141  0.21993549 -0.597700497  0.25486357
##          PC11          PC12
## CRIM     -0.085174472 -0.04657971
## ZN        0.068448851  0.08110771
## INDUS     0.115754811  0.25088024
## CHAS      0.002555635 -0.03571847
## NOX      -0.806923521 -0.04446160
## RM       -0.134810819 -0.04636863
## AGE       0.188185942  0.03925830
## DIS      -0.402053350  0.01799746
## RAD       0.114985116  0.63285900
## TAX       0.220704576 -0.72064510
## PTRATIO -0.214232038 -0.02322379
## LSTAT    -0.021599404 -0.02576718
```

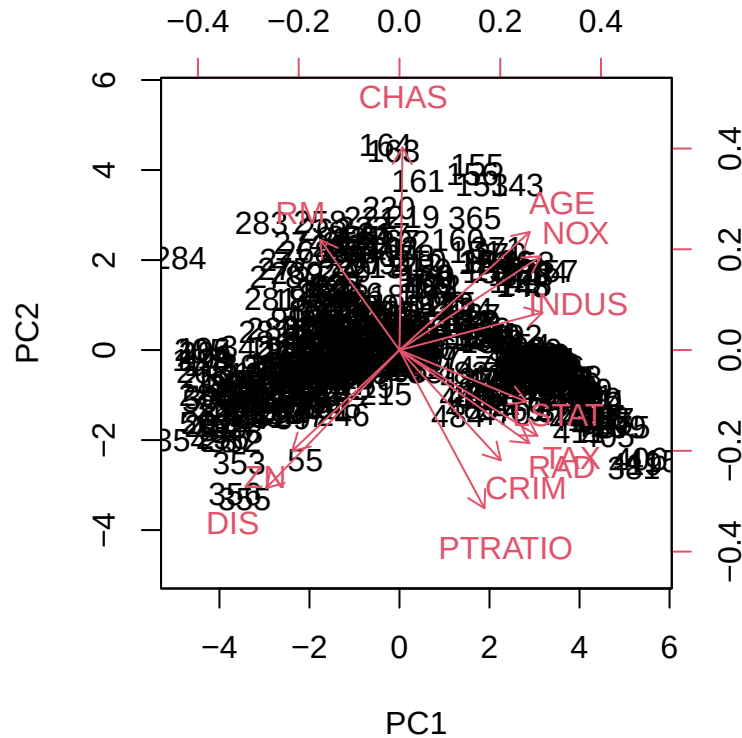
4.3 Principal Component Scores

```
# Principal component scores (rows = observations, cols = PCs)
head(pr.out$x)
```

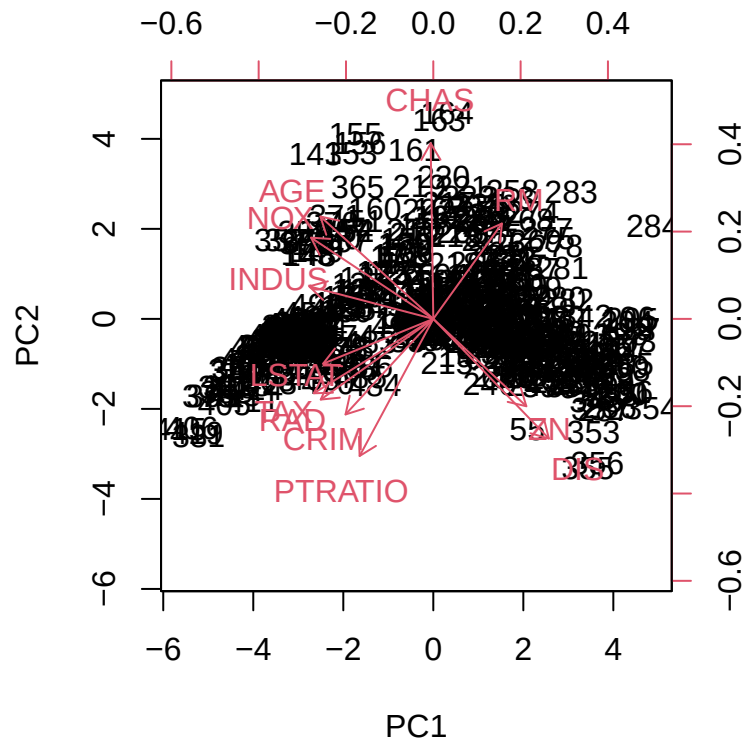
```
##          PC1          PC2          PC3          PC4          PC5          PC6
## [1,] -2.047164  0.861424322  0.08126262  0.7334639  0.7116768 -0.3008453
## [2,] -1.385954  0.433814514 -0.83168088  0.4902902 -0.2168959 -0.5660622
## [3,] -2.029568  0.637119602  0.01543283  0.9561439 -0.8229194 -0.5076444
## [4,] -2.575093 -0.006483661 -0.11127506  0.6203118 -1.1556285 -0.6581153
## [5,] -2.413700  0.094804510 -0.10158521  0.6919878 -1.0766316 -0.7383737
## [6,] -2.168019 -0.133591740 -0.68374944  0.3116309 -0.7801709 -0.6808604
##          PC7          PC8          PC9          PC10          PC11          PC12
## [1,]  0.2968741  0.39096995  0.66921216 -0.01933973 -0.03718345 -0.365330063
## [2,] -0.2235426  0.15974421  0.11120407 -0.64120585  0.10677300  0.071021909
## [3,]  0.1041077 -0.05665561 -0.18767750 -0.48594437 -0.14335885  0.014215868
## [4,]  0.2539674  0.35457470  0.02758278 -0.34356982 -0.52356577 -0.007575941
## [5,] -0.1360873  0.43437152 -0.14892385 -0.38505562 -0.50358929 -0.014516364
## [6,]  0.1084599  0.46791563  0.37312176 -0.50814092 -0.33518349  0.039722735
```

5. Visualize PCA: Biplot

```
# Basic biplot  
biplot(pr.out, scale = 0)
```



```
# Optional: Flip PC1 direction (no impact on interpretation)  
pr.out$rotation[,1] = -pr.out$rotation[,1]  
pr.out$x[,1] = -pr.out$x[,1]  
biplot(pr.out, scale = 0)
```

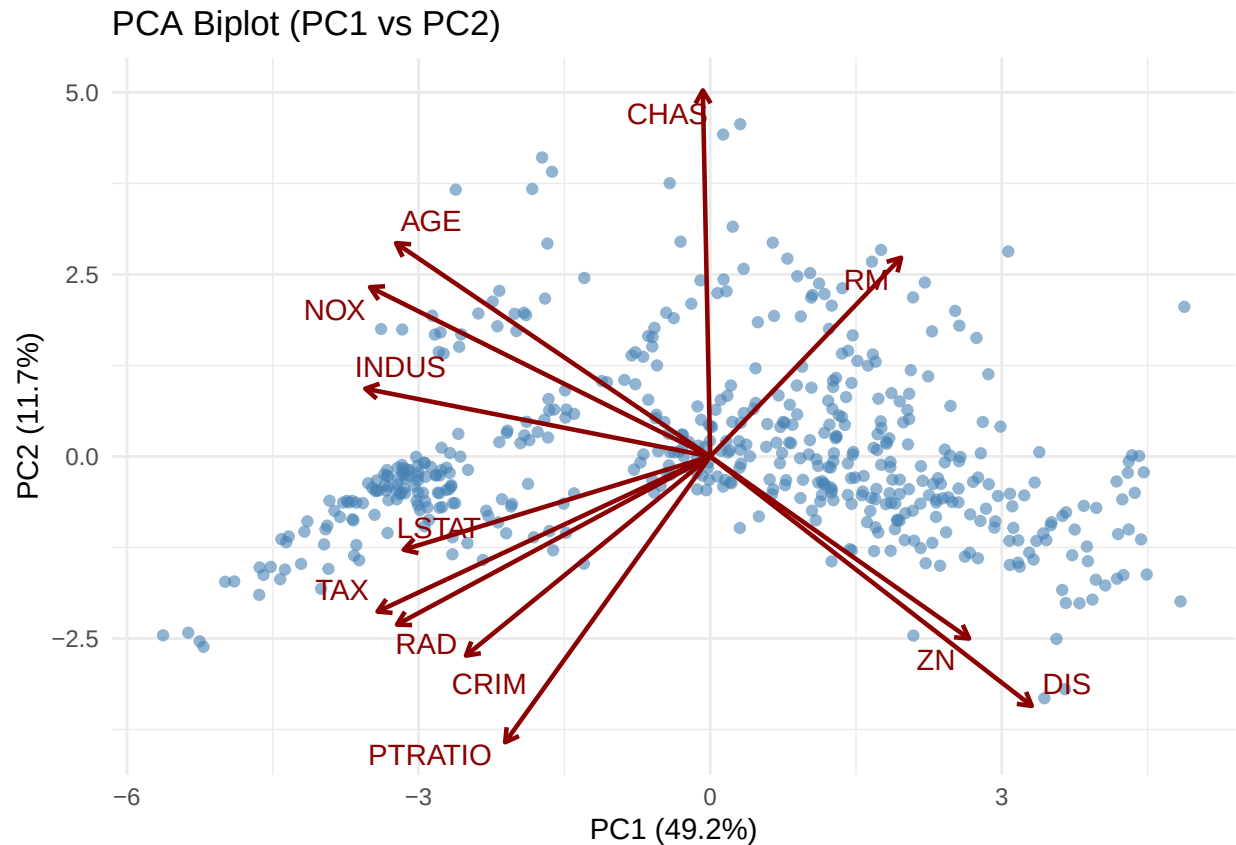


```
# Extract scores and loadings
scores <- as.data.frame(pr.out$x[, 1:2]) # PC1 and PC2
loadings <- as.data.frame(pr.out$rotation[, 1:2])
loadings$varname <- rownames(loadings)

# Scale loadings for visualization (arrows)
loadings_scaled <- loadings
loadings_scaled[, 1:2] <- loadings_scaled[, 1:2] * 10 # adjust scale for better arrow visibility

# Create biplot
p <- ggplot(scores, aes(x = PC1, y = PC2)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_segment(data = loadings_scaled,
    aes(x = 0, y = 0, xend = PC1, yend = PC2),
    arrow = arrow(length = unit(0.2, "cm")),
    color = "darkred", linewidth = 0.8) +
  geom_text_repel(data = loadings_scaled,
    aes(x = PC1, y = PC2, label = varname),
    color = "darkred", size = 4) +
  labs(title = "PCA Biplot (PC1 vs PC2)",
    x = paste0("PC1 (", round(summary(pr.out)$importance[2, 1] * 100, 1), "%)"),
    y = paste0("PC2 (", round(summary(pr.out)$importance[2, 2] * 100, 1), "%)")) +
  theme_minimal()

# Show plot
print(p)
```



```
# Save to file
```

```
ggsave(filename = "../output/pca/pca_biplot.png", plot = p, width = 8, height = 8, dpi = 300, bg="white")
```

6. Variance Explained

```
# Summary of PCA: includes proportion of variance explained
```

```
pca_summary <- summary(pr.out)
```

```
pca_summary
```

```
## Importance of components:
```

```
##          PC1    PC2    PC3    PC4    PC5    PC6    PC7
## Standard deviation  2.4304 1.1832 1.08660 0.9244 0.89574 0.73225 0.62937
## Proportion of Variance 0.4923 0.1167 0.09839 0.0712 0.06686 0.04468 0.03301
## Cumulative Proportion 0.4923 0.6089 0.70732 0.7785 0.84539 0.89007 0.92308
##          PC8    PC9    PC10    PC11    PC12
## Standard deviation  0.52723 0.47451 0.43152 0.41256 0.25204
## Proportion of Variance 0.02316 0.01876 0.01552 0.01418 0.00529
## Cumulative Proportion 0.94624 0.96501 0.98052 0.99471 1.00000
```

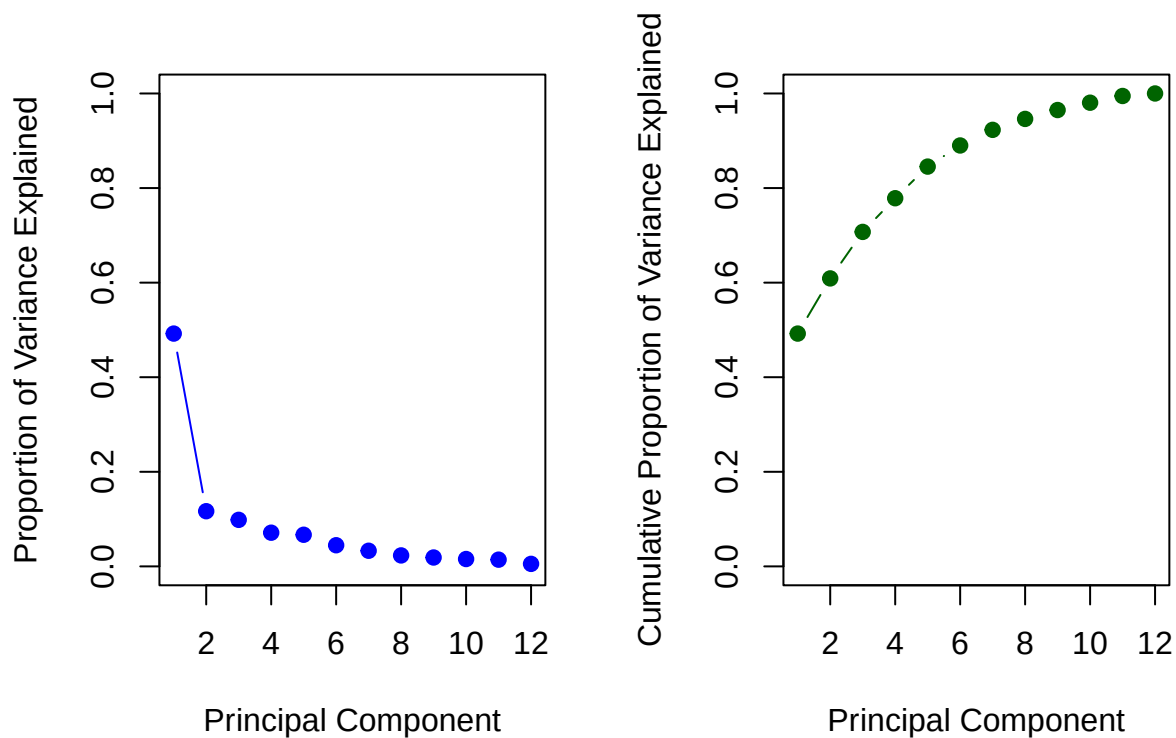
7. Scree Plot & Cumulative Variance

a. Plot

```
# Extract importance matrix
importance <- pca_summary$importance

# Scree plot and cumulative variance
par(mfrow = c(1, 2))
plot(importance[2,], xlab = "Principal Component",
     ylab = "Proportion of Variance Explained", ylim = c(0, 1),
     type = "b", col = "blue", pch = 19)

plot(importance[3,], xlab = "Principal Component",
     ylab = "Cumulative Proportion of Variance Explained",
     ylim = c(0, 1), type = "b", col = "darkgreen", pch = 19)
```



```
par(mfrow = c(1, 1))
```

b. ggplot2 Version

```
# Extract variance explained from PCA summary
pca_summary <- summary(pr.out)
importance <- pca_summary$importance

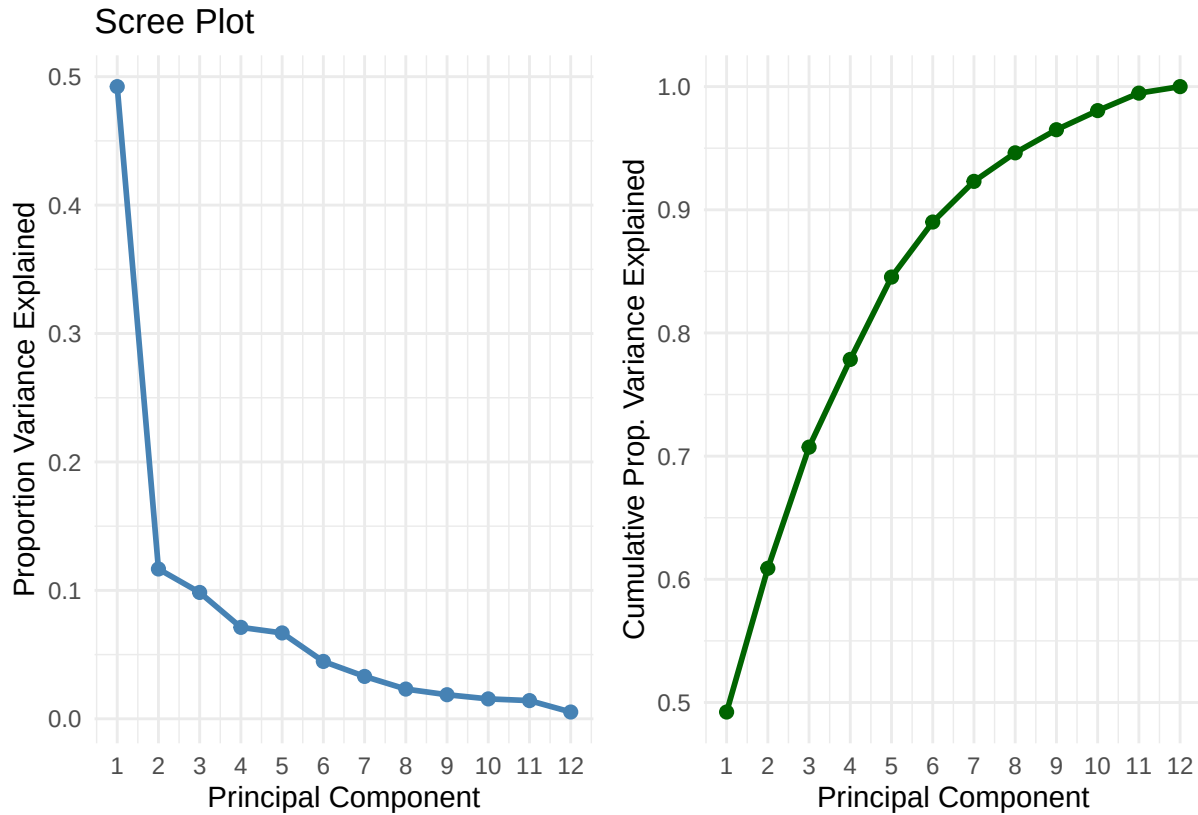
# Create a data frame for plotting
pc_data <- data.frame(
  PC = paste0("PC", 1:ncol(pr.out$rotation)),
  Variance = importance[2, ],
  Cumulative = importance[3, ],
  PC_num = 1:ncol(pr.out$rotation)
)

# Scree Plot
scree_plot <- ggplot(pc_data, aes(x = PC_num, y = Variance)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "steelblue", size = 2) +
  scale_x_continuous(breaks = pc_data$PC_num) +
  labs(title = "Scree Plot",
       x = "Principal Component",
       y = "Proportion Variance Explained") +
  theme_minimal()

# Cumulative Variance Plot
cum_plot <- ggplot(pc_data, aes(x = PC_num, y = Cumulative)) +
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "darkgreen", size = 2) +
  scale_x_continuous(breaks = pc_data$PC_num) +
  labs(x = "Principal Component",
       y = "Cumulative Prop. Variance Explained") +
  theme_minimal()

# Combine two plots side-by-side using patchwork
combined_plot <- scree_plot + cum_plot

print(combined_plot)
```



```
# Save the combined plot to file
ggsave("../output/pca/pca_scree_and_cumulative_pve.png",
        plot = combined_plot, width = 12, height = 6, dpi = 300)
```

8. Interpretation

- The **loadings** (rotation) show how each original variable contributes to the PCs.
- The **principal component scores** (x) are new features that can be used for clustering.
- From the **variance explained**, you can decide how many components to retain (e.g., enough to capture 90% of the variance).
- The **biplot** helps visualize the relationship between variables and observations in the reduced PC space.